Eugene Yakhnenko

Staff Software Engineer

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SUMMARY

Software engineer with 11+ years of experience delivering products end-to-end. Track record of driving org-wide architectural decisions and building developer tools and design systems that multiply team velocity. Deep hands-on range from React and TypeScript to Go, PostgreSQL, and cryptographic systems.

EXPERIENCE

Staff Software Engineer — Feb 2025 - Present

Virtru

- Set the technical architecture for the platform's web admin UI—a customer-facing surface used by enterprise customers to manage their deployments—accelerating development and eliminating duplication across teams, and extending the product beyond CLI-only administration
- Built an Electron-based tool pairing a chat UI with a live browser preview powered by Playwright, enabling non-technical roles (marketing, product) to produce PoCs and products through an AI agent that can see, interact with, and modify a web application in real time
- Directed cross-team monorepo upgrade (65+ packages, 7+ teams) from Node 18 to 22 and pnpm 8 to 10, moving company OKR from "at risk" to "on track."
- Developed proof-of-concept for ABAC-driven cryptographic video redaction with segment-level access control over encrypted video streams, validating technical feasibility of a novel approach and transforming an Engineering Day initiative into a product epic
- Eliminated duplicated authentication implementations across all frontend applications by identifying a systemic pattern, authoring an ADR evaluating standardization options, and defining the organization's OIDC standard
- Identified pre-1.0 API design debt in the OpenTDF open-source WebSDK—two divergent auth patterns with hidden ordering dependencies and silent failures—drove architectural alignment and executed the migration, eliminating an entire class of integration errors
- Delivered company-wide knowledge shares on adaptive video streaming with end-to-end encryption, and security implications of AI-assisted development, presenting the Three Layers of Correctness framework for evaluating AI-assisted development quality
- Improved AI-assisted development for consumers of the OpenTDF open-source platform by defining its AI-consumable knowledge layer (llms.txt), consolidating context across four repositories and enabling an estimated 2–3x faster integration with fewer architectural missteps
- Created a lightweight, framework-agnostic agentic workflow orchestration library for chaining LLM agent steps. Workflows are defined as data, not code—the library handles sequencing, parallelism, routing, retries, abort, and interactive turns while the consuming app provides the LLM runtime

Senior Software Engineer — July 2021 - Feb 2025

Underline

- Implemented a customer-facing checkout platform that reduced onboarding time and significantly enhanced the user experience, contributing to acceleration in business growth alongside other initiatives
- Owned technical design and delivery of a core internal platform adopted by Marketing, Operations, and Support, defining the architecture and onboarding three teams onto the system
- Drove adoption of domain-driven design across the frontend codebase, establishing bounded contexts that aligned team ownership boundaries with business domains and reduced cross-team coupling
- Built a hybrid mobile app using React, Vite and Capacitor to allow network technicians perform internet installation processes
- Designed and implemented PostgreSQL database schema updates and utilized Go and SQLBoiler to streamline database interactions for critical client-facing features
- Developed a robust suite of acceptance and integration tests using Playwright and QUnit for the checkout platform, supporting a smooth migration to a new catalog and billing system

Senior Frontend Engineer **Apr 2020 - July 2021**

BairesDev

- Redesigned and implemented a web application in Angular to assess candidates' technical skills, improving talent acquisition
- Migrated a legacy time tracking application, implementing a modern React-based frontend and a C# .NET backend
- Implemented a live coding tournament using WebSockets, accelerating talent acquisition through real-time assessment
- Established an AWS-based microservices deployment strategy over an existing monolith architecture, utilizing Docker and load balancers for scalability and efficiency
- Mentored junior developers and led adoption of coding standards and CI linting, reducing friction in code reviews

Frontend Engineer **Feb 2018 - Apr 2020**

TicketNetwork

- Initiated the migration from AngularJS to Angular 8 for a legacy application, resulting in a 3x improvement in rendering performance and a more maintainable codebase
- Created an interactive map for filtering venue tickets, enabling brokers to manage inventory in real-time
- Upgraded an existing web app to a PWA, integrating service workers and manifest files

Frontend Engineer **Apr 2015 - Jan 2018**

8a Marketing

- Developed responsive web pages, hybrid mobile apps, web applications, and integrations with various APIs
- Built multiple CMS platforms in PHP enabling clients to manage their content

EDUCATION

EEST3 Mar del Plata, Argentina | **2005** | *Associates in Processes and Production*

PROJECTS

Autentico | *github.com/eugenioenko/autentico* | 41 stars

- Built a self-hosted OAuth 2.0 / OpenID Connect Identity Provider in Go—single binary, zero external dependencies, backed by SQLite. Handles login, MFA, passkeys, session management, and token issuance
- Implemented standards-compliant Authorization Code + PKCE, RS256-signed JWTs, WebAuthn/passkeys, TOTP, OIDC Discovery, and token introspection—verified through 1,800+ tests and RFC compliance audits
- Ships with a built-in React admin dashboard compiled into the binary and a live demo platform that provisions ephemeral instances on demand

TTT | *github.com/eugenioenko/ttt* | 941 stars

- Built a full-featured terminal IDE in Go as a real alternative to VS Code and Zed—single binary, zero config. Features syntax highlighting, multi-cursor editing, LSP-powered completions, diagnostics, and an integrated terminal emulator

Yarcl | *github.com/eugenioenko/yarcl*

- Built a React design system designer where a single TypeScript config defines design tokens, typed component props, and build-time CSS. Supports custom component recipes, light/dark themes, and a live theme playground

Skedoodle | *github.com/eugenioenko/skedoodle*

- Built a real-time collaborative whiteboard using event sourcing and WebSockets—every stroke is an immutable command in an append-only log, with the canvas as a projection. Achieves 0% CPU at idle via fully event-driven rendering with incremental dirty-region redraws and viewport culling

Kasper-js | *github.com/eugenioenko/kasper-js*

- Built a lightweight component framework with signal-based reactivity, no virtual DOM, and no build step required—16KB script tag delivers components, router, slots, and lazy loading. Custom recursive-descent template parser works under strict CSP. Under 4,000 lines of TypeScript with 95% test coverage